

EDUCATION

The University of British Columbia
Ph.D. in Statistics Sep 2013 - Feb 2018
Thesis: Scalable sequential Monte Carlo methods and probabilistic approach to combinatorial problems
Advisors: Alexandre Bouchard-Côté, Ph.D. and James V. Zidek, Ph.D., FRSC

MSc. in Statistics Sep 2011 - Aug 2013
Thesis: Entangled Monte Carlo
Advisor: Alexandre Bouchard-Côté, Ph.D.

University of Waterloo
Bachelor of Mathematics, Honours, Co-op Jan 2004 - May 2009
Major: Computer Science
Minor: Combinatorics and Optimization

EMPLOYMENT

University of Rochester Medical Center
Assistant Professor Jan 2024 - Current
Department of Biostatistics and Computational Biology

University of Rochester Medical Center
Research Assistant Professor Oct 2022 - Dec 2023
Department of Biostatistics and Computational Biology

Fred Hutchinson Cancer Research Center
Postdoctoral Research Fellow Jan 2020 - Sep 2022
Computational Biology Program, Public Health Sciences Division
Mentors: Frederick Matsen, Ph.D. and Raphael Gottardo Ph.D.

Science for Life Laboratory, KTH Royal Institute of Technology
NSERC Postdoctoral Research Fellow Nov 2017 - Nov 2019
Department of EECS
Mentor: Jens Lagergren, Ph.D.

MANUSCRIPTS UNDER PREPARATION

1. **S-H. Jun**, M. K. Halushka, and M. N. McCall. NBSR: a Negative Binomial Softmax Regression model for microRNA-seq data analysis. Under review at *Biostatistics*.
2. O. Kviman, **S-H. Jun**, and J. Lagergren. LN's t-Test: A Principled Approach to t-Testing in Single-Cell RNA Sequencing. <https://doi.org/10.1101/2025.03.12.642799>. In preparation.

REFEREED PUBLICATIONS

13. Y. Zheng, D. P. Caron, J.Y. Kim, **S-H. Jun**, Y. Tian, F. Mair, K. D. Stuart, P. A. Sims, R. Gottardo. ADTnorm: Robust integration of single-cell protein measurement across CITE-seq datasets. *Nature Communications*. Accepted.
12. A. M. Baran, A. H. Patil, E. Aparicio-Puerta, **S-H. Jun**, M. K. Halushka, M. N. McCall. miRglm: a generalized linear mixed model of isomiR-level

- counts improves estimation of miRNA-level differential expression and uncovers variable differential expression between isomiRs. *Genome Biol.* 26, 102 (2025).
11. H. Koptagel, **S-H. Jun**, J. Hård, J. Lagergren. Scuphr: A probabilistic framework for cell lineage tree reconstruction. *PLoS Comput. Biol.* 20, e1012094 (2024).
 10. **S-H. Jun**, H. Nasif, C. Jennings-Shaffer, D. H. Rich, A. Kooperberg, M. Fourment, C. Zhang, M. A. Suchard, F. A. Matsen IV. A topology-marginal composite likelihood via a generalized phylogenetic pruning algorithm. *Algorithms Mol. Biol.* 18, 10 (2023).
 9. D. A. Oyong, F. J. Duffy, M. L. Neal, Y. Du, J. Carnes, K. V. Schwedhelm, N. Hertoghs, **S-H. Jun**, H. Miller, J. D. Aitchison, S. C. De Rosa, E. W. Newell, M. J. McElrath, S. M. McDermott, and K. D. Stuart. Distinct immune responses associated with vaccination status and protection outcomes after malaria challenge. *PLoS Pathog.* 19, e1011051 (2023).
 8. **S-H. Jun**, H. Toosi, J. Mold, C. Engblom, X. Chen, C. O’Flanagan, M. Hagemann-Jensen, R. Sandberg, S. Aparicio, J. Hartman, A. Roth, J. Lagergren. Reconstructing clonal tree for phylo-phenotypic characterization of cancer using single-cell transcriptomics. *Nature Communications.* 14, 982 (2023).
 7. X. Chen, E.G. Sifakis, S. Robertson, S.Y. Neo, **S-H. Jun**, J. Lötvrot, V. Jovic, J. Bergh, T. Foukakis, J. Lagergren, A. Lundqvist, R. Ma, and J. Hartman. Breast cancer patient-derived whole-tumor cell culture model for efficient drug profiling and treatment response prediction. *Proceedings of the National Academy of Sciences of the United States of America.* 120, e2209856120 (2023).
 6. M. M. Neyshabouri, **S-H. Jun**, and J. Lagergren. Inferring tumor progression in large datasets. *PLoS Comput. Biol.* 16(10), p.e1008183 (2020).
 5. **S-H. Jun**, S. Wong, J. Zidek, and A. Bouchard-Côté. Sequential decision model for inference and prediction on non-uniform hypergraphs with application to knot matching from computational forestry. *The Annals of Applied Statistics.* 13(3), pp. 1678-1707 (2019).
 4. E. Haber, L. Ruthotto, E. Holtham, **S-H. Jun**. Learning across scales - A multiscale method for convolution neural networks. *Association for the Advancement of Artificial Intelligence (AAAI).* (2018).
Acceptance rate: 933/3800.
 3. **S-H. Jun**, A. Bouchard-Côté, S. Wong, and J. Zidek. Sequential graph matching with sequential Monte Carlo. *International Conference on Artificial Intelligence and Statistics (AISTATS).* pp. 1075–1084 (2017).
Acceptance rate: 168/530.
 2. **S-H. Jun** and A. Bouchard-Côté. Memory (and time) efficient sequential Monte Carlo. *International Conference on Machine Learning (ICML).* pp. 514–522 (2014).
Acceptance rate: 310/1238.
 1. **S-H. Jun**, L. Wang, and A. Bouchard-Côté. Entangled Monte Carlo. *Advances in Neural Information Processing Systems 25 (NIPS).* pp. 2735–2743 (2012).
Acceptance rate: 370/1467. Spotlight talk: 72/1467.

TEACHING EXPERIENCES

Spring 2025 – BST 570: Probabilistic Machine Learning *Instructor*

Topics: Graphical models, hidden Markov models, message passing algorithm, Markov chain Monte Carlo methods, sequential Monte Carlo methods, variational inference.
Course website: https://junseonghwan.github.io/teaching/pml_spring25/.

Spring 2024 – BST 434: Genomics Data Analysis

Co-instructor

Topics: bulk and single-cell RNA expression analysis methods, batch effect correction and normalization, microbiome and phylogenetic analysis.

Spring 2023 – IND 419: Introduction to Quantitative Biology

Guest lecture

Topics: R graphics using ggplot2 and differential gene expression analysis.

Fall 2018 – KTH DD2447: Statistical methods in applied computer science

Guest lecture

Topics: importance sampling, sequential Monte Carlo, and particle MCMC methods.

2016-2017 – UBC Master of Data Science program

Academic assistant

Developed assignments and lab materials for newly launched master program. Example of topics covered:

- Analyzing Google N-grams using Map-Reduce on Amazon Web Services (AWS).
- Training deep neural network with Tensorflow using GPU instances on AWS.
- Designing an A/B testing for web interface using R Shiny.

Fall 2015 – UBC STAT 300: Intermediate statistics for applications

Head teaching assistant

Topics: Non-parametric tests including Kruskal-Wallis, permutation test, and Fisher's exact test.

Winter 2014 – UBC STAT 547: Statistical modelling with stochastic processes

Teaching assistant

Topics: Selected topics in non-parametric Bayesian methods, continuous time Markov processes, point processes.

Summer and Fall 2012 – UBC STAT 447B/547B: Methods for statistical learning

Course developer and teaching assistant

Developed course materials on Boosting, generalized additive models, splines, regression trees and random forest, LASSO, K-NN classifier.

Winter 2012 – UBC STAT 441 Multivariate statistical methods

Teaching assistant

Topics: Multivariate hypothesis testing and ANOVA, PCA, latent variable analysis, and discriminant analysis.

Fall 2011 – UBC STAT 203: Statistical methods

Head teaching assistant

Introduction to statistics including central limit theorem and hypothesis testing.

PRESENTATIONS

Rutgers New Jersey Medical School

Oct, 2024

Department of Pharmacology, Physiology, and Neuroscience. Newark, NJ, USA

Talk title: Bridging statistics, machine learning, and AI: Uncovering hidden mechanisms in biomedical research.

Science for Life Laboratory

Oct, 2024

Solna, Sweden

Talk title: Improved differential expression analysis of miRNA-seq data by modeling competition to be counted.

Bioconductor Conference July, 2024

Grand Rapids, MI, USA

Talk title: Improved differential expression analysis of miRNA-seq data by modeling competition to be counted.

Research in Computational Molecular Biology (RECOMB) Apr, 2024

Boston, MA, USA

Poster presentation: Statistical modeling of microRNA-sequencing data.

The Classification Society Annual Meeting June, 2023

Rochester, NY, USA

Oral presentation: Reconstructing cancer evolution as a Bayesian co-clustering problem.

PhyloMania Nov, 2020

Virtual conference.

Oral presentation: Generalized phylogenetic pruning algorithm.

Probabilistic modelling in genomics Nov, 2018

Cold Spring Harbor, NY, USA

Poster presentation: Reconstruction of tumor phylogeny from single-cells via joint probabilistic analysis of bulk DNA and scRNA-seq.

Conference on machine and other intelligence Sep, 2018

Norrköping, Sweden

Poster and oral presentation: Large scale machine learning for the single cell revolution.

International Conference on AI and Statistics (AISTATS) Oct, 2017

Fort Lauderdale, FL, USA

Poster presentation: Sequential graph matching with sequential Monte Carlo.

Joint Statistical Meeting (JSM) Aug, 2017

Baltimore, MD, USA

Poster presentation: Sequential graph matching and streaming sequential Monte Carlo.

International Conference on Machine Learning July, 2014

Beijing, China

Oral presentation: Memory (and time) efficient sequential Monte Carlo.

Randomized Algorithm Workshop at NeurIPS Dec, 2013

Lake Tahoe, NV, USA

Poster presentation: Using a stochastic map view of sequential Monte Carlo for memory and network efficiency.

Annual Meeting of the Statistical Society of Canada May, 2013

Edmonton, AB, Canada

Poster and oral presentation: Exploring spatial and temporal heterogeneity of environmental noise in Toronto.

Winner of the case study competition.

NeurIPS Dec, 2012
Lake Tahoe, NV, USA
Spotlight talk and poster presentation: Entangled Monte Carlo.

UBC-SFU joint seminar Sep, 2012
Vancouver, BC, Canada
Oral presentation: Sequential importance sampling and bootstrap particle filter.

**COMMITTEE
SERVICES**

- Hannah Swan. Ph.D. in Statistics. URMCI.
- Jonathan Klus. Ph.D. in Statistics. URMCI.
- Josh Geiger. M.S. in Biostatistics (2024). URMCI.

**PROFESSIONAL
SERVICES**

Reviewer

- Advances in Approximate Bayesian Inference (AABI)
- Bayesian Analysis
- British Journal of Cancer
- BMC Bioinformatics
- International Conference on Research In Computational Molecular Biology (RECOMB).
- International Conference on Artificial Intelligence and Statistics (AISTATS)
- International Conference on Machine Learning (ICML)
- International Conference on Learning Representations (ICLR)
- Neural Information Processing Systems (NeurIPS)

PhD Dissertation Award Committee 2025
University of Rochester Medical Center

Manager of Statistical consulting services 2016 - 2017
Department of Statistics, University of British Columbia
- Developed operating guidelines for the consulting services.

Senior consultant 2015 - 2017
Department of Statistics, University of British Columbia
- Provided statistical advice to graduate students and postdoctoral researchers.
- Recruited and mentored junior consultants.

Graduate student seminar organizer 2014 - 2016
Department of Statistics, University of British Columbia
- Invited speakers for weekly seminar.
- Organized lecture series on parallel computing in R, statistical analysis of network data, deep neural networks, and sports analytics.

Academic guide 2013 - 2014, 2015 - 2016
International Graduate Student Preparation Program, University of British Columbia
- Provide mentorship to prospective graduate students. Cultivate research interests and provide guidance on research statements.

**AWARDS &
GRANTS**

2024-25	\$30,000 USD	GIDS + AI Pilot Grant with Drs. McCall and Rossman (URMC)
2024-25	\$35,000 USD	CTSI NBEM Pilot Grant (URMC)
2018-20	\$90,000 CAD	NSERC Postdoctoral Fellowship
2017	\$850 CAD	CRM Industrial Problem Workshop Travel Award
2017	\$1,000 USD	AISTATS Travel Award
2013-17	\$18,000 CAD	Faculty of Science Graduate Award (Ph.D)
2014	\$500 USD	ICML Travel Award
2013	\$500 CAD	SSC Case Study Competition Winner
2011-13	\$1,000 CAD	Faculty of Science Graduate Award (MSc.)
2012	\$400 USD	NIPS Travel Award
2011	\$5,500 CAD	NSERC Undergraduate Student Research Award